

PAF9615C2: Ultra Small FIR Thermopile Sensor

General Description

The PAF9615C2 is a low-power infrared (IR) thermopile sensor with integrated temperature computation function. It measures a target object's temperature (T_O) without physical contact the object. The object temperature is converted from the voltage of the thermopile sensor and ambient temperature (T_A). Both temperatures will be calibrated with built-in calibration coefficients which stored in the internal memory. It supports four output modes of conversion state based on application requirements. This is a low power consumption thermopile sensor ideal for battery-operated cordless applications.

Key Features

- 16-bit object temperature data length
 - *Accuracy $\pm 0.5^{\circ}\text{C}$ @ 10 to 45°C (T_O and T_A)
 - Resolution : 0.03125°C
 - FOV : 50°
- 16-bit ambient temperature data length
 - Accuracy $\pm 0.5^{\circ}\text{C}$ @ 10°C to 45°C
 - Resolution : 0.03125°C
- Integrated temperature computation function
 - Direct output object temperature
 - Programmable alert temperature
 - Built-in memory for calibration coefficients
- IO interface
 - Support two-wire I²C, 400kbps
 - Support two programmable I²C addresses
- Embedded Emissivity and Decay compensation
- Low power design

Applications

- Wearable device temperature sensing
- Non-Contact Object temperature sensing
- Skin temperature monitor to prevent heat stroke
- Ambient temperature detection
- Smart Appliance-Non-contact temperature monitor
- Temperature feedback control application.

Key Parameters

Parameters	Value
Operating Temperature	-20 to 85°C
Object Temperature, T_O	Measurement Range: -20 to 380°C *Accuracy $\pm 0.5^{\circ}\text{C}$ @ T_O : 10 to 45°C and T_A : 10 to 45°C
Ambient Temperature, T_A	Measurement Range : -20 to 85°C Accuracy : $\pm 0.5^{\circ}\text{C}$ @ 10 to 45°C
Supply Voltage	VDDMA: 2.2 to 3.6 V VDDIO: 1.62 to 3.6 V
Typical Current Consumption @ 3.3V, Report Rate = 4Hz	Operation: $225\mu\text{A}$ Suspend: $2\mu\text{A}$ Shutdown: $2\mu\text{A}$ Note: pure sensor current without I/O toggling. Normal mode @ $t_{\text{CONVERSION}} < t_{\text{REPORT}}$
Package Size	$3.3 \times 3.3 \times 1.53 \text{ mm}^3$

Note: * The higher accuracy could be tuned to optimize for application requiring narrow object temperature range (such as body temperature). For more information, do approach PixArt personnel.

Ordering Information

Part Number	Description	Package Type	Packing Type	MOQ
PAF9615C2	Thermopile Sensor	Ceramic	Tray	TBD



For any additional inquiries, please contact us at <https://www.pixart.com>

PAF9615C2 I²C Interface Reference Control Flow

Ordering Information

Part Number	Package Type
PAF9615C2	Ceramic 3.3mm X 3.3mm X 1.53mm



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1.0 Introduction

The PAF9615C2 is a low-power infrared (IR) thermopile sensor with integrated temperature computation function. It measures a target object's temperature (T_O) without direct contact the object. The object temperature is converted from the voltage of the thermopile sensor and ambient temperature (T_A). Both temperatures will be calibrated with built-in calibration coefficients which stored in the internal memory. The communication IO interface is two-wire I²C with speed up to 400kbps in the Fast-mode. PAF9615C2 supports wide temperature range measurement from -20 to 380°C with 80° FOV.

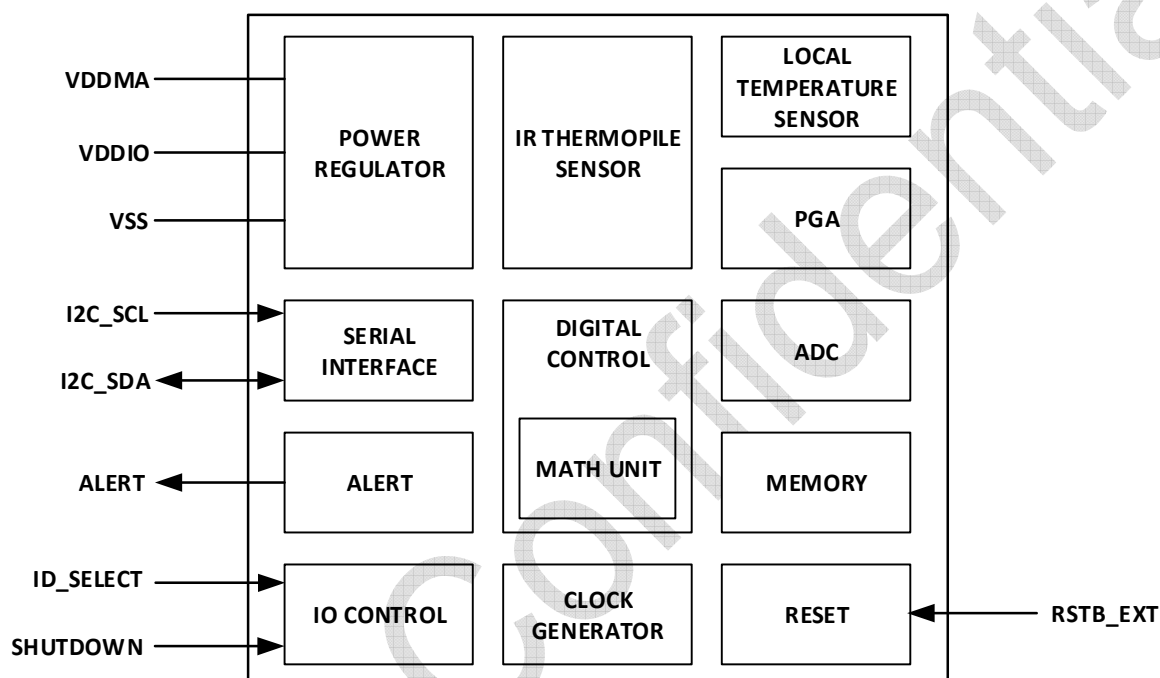


Figure 1. Block Diagram

1.1 Terminology

Term	Description
T_A	Ambient temperature
T_O	Object temperature
FOV	Field of View
SNR	Signal to noise ratio

1.2 Reference design

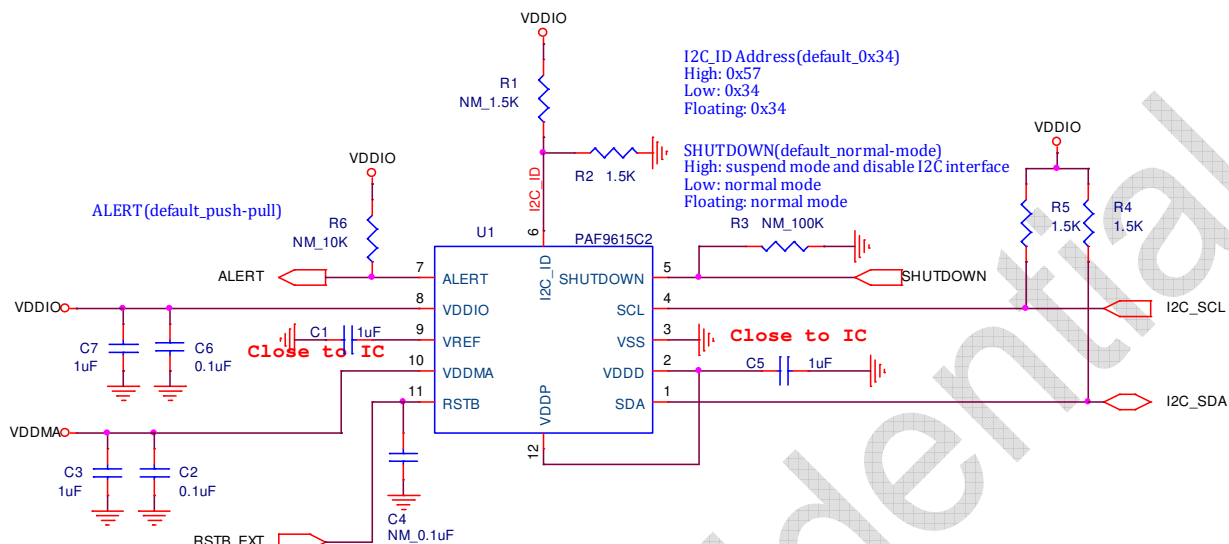


Figure 2. Reference Schematic

Note: "NM" is reserved component.

1.3 Pins Description

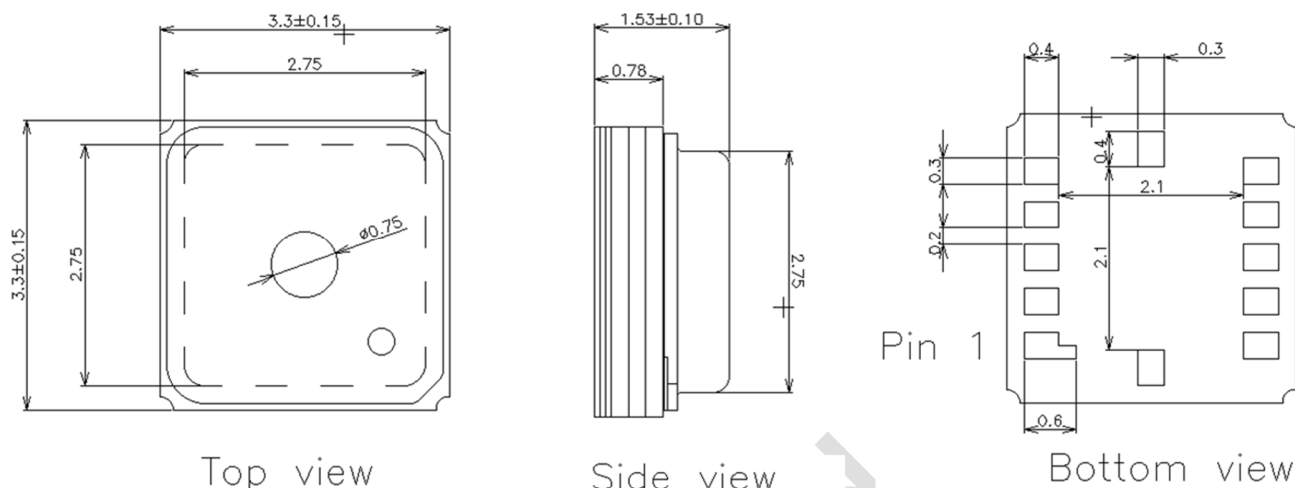


Figure 3. Module Pin Configuration

Table 1. Signal Pins Description

Function	Pin No.	Signal Name	Type	Description
Power Supplies	8	VDDIO	Input	I/O power supply (1.62 to 3.6V). Connect 1 μ F and 0.1 μ F capacitor to Ground
	2	VDDD	Output	Digital power supply. Connect 1 μ F capacitor to Ground
	3	VSS	Ground	Ground
	9	VREF	Output	Analog reference power. Connect 1 μ F capacitor to Ground
	10	VDDMA	Input	Main power supply (2.2 to 3.6V) for internal power regulator. Connect 1 μ F and 0.1 μ F capacitor to Ground
	12	VDDP	Input	Integrated memory power. Connect to VDDD
Interface	4	I2C_SCL	Input	I ² C clock pin
	1	I2C_SDA	I/O	I ² C bidirectional data pin
Functional I/O	7	ALERT	Output	Alert output pin; active low It can be programmed register 0x2F[0] as a push-pull or open drain. Push-pull is default
	11	RSTB_EXT	Input	Hardware reset pin; active low Floating when not in use
	5	SHUTDOWN	Input	Shutdown pin; active high High: sensor suspend mode and disable I ² C interface Low/ Floating: sensor normal operation
	6	I2C_ID	Input	I ² C 7bit address selection pin High: I ² C ID = 0x57 Low/ Floating: I ² C ID = 0x34

1.4 Package and PCB footprint

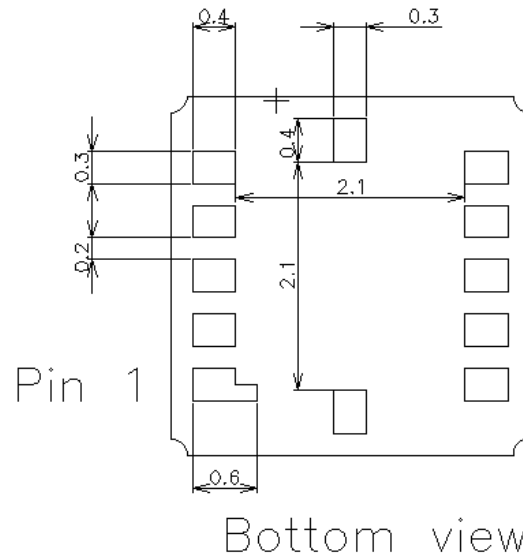


Figure 4. PCB Footprint

2.0 I²C Serial Interface Communication

2.1 Operation Protocol

The protocol is derived from the three above-mentioned packet formats. Below is the combination of the packet format in different operations.

Single Write:

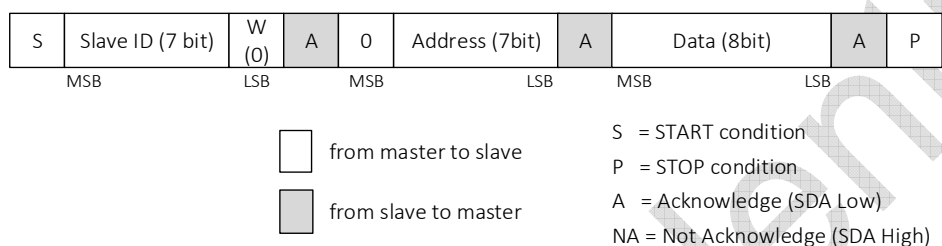


Figure 5. Single Write

1. The host initiates a START condition with Slave ID and LSB bit '0' as write.
2. If match Slave ID address, the sensor responds an ACK.
3. The host sends register address.
4. The sensor responds an ACK when receiving address command.
5. The host sends data to the sensor.
6. The sensor write data to register and responds an ACK.
7. The host terminate the operation with a STOP condition.

Single Read:

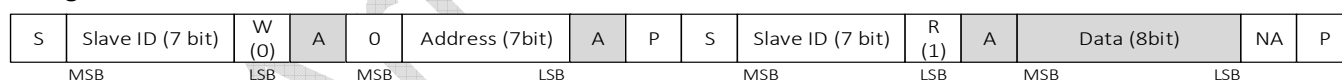


Figure 6. Single Read

8. The host initiates a START condition with Slave ID and LSB bit '0' as write.
9. If match Slave ID address, the sensor responds an ACK.
10. The host sends register address.
11. The sensor responds an ACK upon receiving address command.
12. The host terminate the operation with a STOP condition.
13. The host reinitiates a START condition with Slave ID and LSB bit '1' as read.
14. If match Slave ID address, the sensor responds an ACK.
15. The sensor sends register data to the host.

16. The host responds a NACK and terminate the operation with a STOP condition.

▪ Burst Read with Consecutive Addressing:

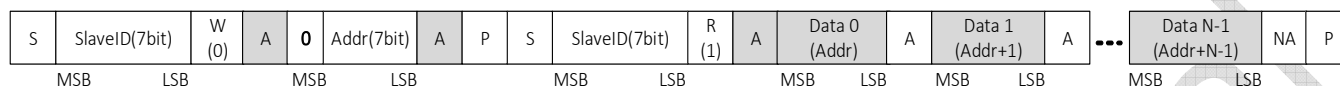


Figure 7. Burst Read with Consecutive Addresses

When the address mode is 0, the register address will auto increase by 1 and output data in sequence:

17. The host initiates a START condition with Slave ID and LSB bit '0' as write.
18. If match Slave ID address, the sensor responds an ACK.
19. The host sends register address with MSB bit '0' in address mode bit.
20. The sensor responds an ACK upon receiving address command.
21. The host terminate the operation with a STOP condition.
22. The host reinitiates a START condition with Slave ID packet LSB bit '1' as read.
23. If match Slave ID address, the sensor responds an ACK.
24. The sensor sends register data to the host with register address.
25. The host continue issue an ACK for next register data read request with register address incremental by 1.
26. The host repeat step 9 until a NACK is issued.
27. The host terminate the operation with a STOP condition.

3.0 Operating Specifications

3.1 Absolute Maximum Ratings

Table 2. Absolute Maximum Ratings

Parameters	Symbol	Min.	Max.	Unit	Notes
Storage Temperature	T_S	-25	85	°C	
Lead-free Solder Temperature	T_P	-	245	°C	
Main Power Voltage	V_{DDMA_MAX}	-0.4	3.6	V	
I/O Voltage	V_{DDIO_MAX}	-0.4	3.6	V	
I/O Pin Input High Voltage	V_{DDIO_IN}	-0.4	$V_{DDIO} + 0.3$	V	All I/O pins
Relative Humidity	RH	0	75	%	Non-condensing, Non-biased
ESD	ESD_{HBM}	-	2	kV	Class 2 on all pins. Human body model, JESD22-A114E.

Notes:

28. At room temperature.

29. Maximum ratings are those values beyond which damage to the device may occur.

30. Exposure to these conditions or conditions beyond those indicated may adversely affect device reliability.

31. Do not intend to operate at the absolute maximum-rated conditions and always restrict to the recommended operating conditions.

3.2 Recommended Operating Conditions

Table 3. Recommended Operating Conditions

Description	Symbol	Min.	Typ.	Max.	Unit	Notes
Operating Temperature	T_{OP}	-20	-	85	°C	
Object Temperature	T_O	-20	-	380	°C	
Ambient Temperature	T_A	-20	-	85	°C	
Power Supply Voltage	V_{DDMA}	2.2	3.3	3.6	V	Power regulator input supply. Includes ripples
Digital Supply Voltage	V_{DDD}	1.62	1.8	1.98	V	Internal regulator output. Includes ripples
I/O Supply Voltage	V_{DDIO}	1.62	3.3	3.6	V	Includes ripples
I ² C Speed	SCL_I2C	-	-	400	kHz	Max value for Fast mode

Note: PixArt does not guarantee the performance if the operating temperature is beyond the specified limit.

3.3 DC Characteristics

Table 4. DC Electrical Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
Power Consumption						
Supply Current @ Suspend	$I_{DDMA-SUSPEND}$	-	2	-	μA	For Sensor only Suspend without I ² C signal toggling
Supply Current @ Operation	$I_{DDMA-OPERATION}$	-	225	-	μA	$t_{CONVERSION} < t_{REPORT}$ T_o Number = 2^{12} T_a Number = 2^9 Report Rate = 4Hz
Peak Power Supply Current	I_{DDMA_MAX}	-	-	255	μA	For VDDMA @ $t_{CONVERSION} < t_{REPORT}$
I/O						
Peak I/O Supply Current	I_{DDIO_MAX}	-	0.1	-	μA	For VDDIO @ I ² C 400kHz
Input High Voltage	V_{IH}	$0.7 \times V_{DDIO}$	-	-	V	
Input Low Voltage	V_{IL}	-	-	$0.3 \times V_{DDIO}$	V	
Output High Voltage	V_{OH}	$V_{DDIO} - 0.4$	-	$V_{DDIO} + 0.4$	V	
Output Low Voltage	V_{OL}	-0.4	-	0.4	V	

Notes: All the parameters are tested under operating conditions: VDDMA = 3.3V, VDDIO = 1.8 and 3.3V, $T_A = 25^\circ C$

3.4 AC Characteristics

Table 5. AC Electrical Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
Power up from VDDMA Rising	t_{PU}	120	-	-	ms	From VDDMA rising to valid interface communication

Notes: All the parameters are tested under operating conditions: $T_A = 25^\circ C$, VDDMA = 3.3V, VDDIO = 3.3V for 3.3V IO application and VDDIO = 1.8 V for 1.8V IO application.

3.5 Accuracy Specification

All the sensor unit is calibrated before delivery. Both calibration and thermal transient coefficient data are stored in integrated memory.

Table 6. Accuracy Specifications

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
T _{object} (T _O) Accuracy			*±0.5°C		°C	T _O =10 to 45 °C and T _A =10 to 45 °C
T _{ambient} (T _A) Accuracy			±0.5°C		°C	T _A =10 to 45 °C
			±1°C		°C	T _A =0 to 70 °C

The sensor can be optimized to support specific application through compromising between object temperature range and accuracy requirement. (For example, human body temperature.). For further enquiry, do approach PixArt personnel.

The [Table 6](#) specified accuracy are under below condition.

- Under the isothermal conditions
- At V_{DDMA} = 3.3V
- The target object needs to be larger than FOV
- The emissivity of the object is 0.98
- T_A and T_O sampling numbers are set to 512 and 4096,

3.6 Field of View

Table 7. Field of View

Parameters	Symbol	Min.	Typ.	Max.	Unit	Conditions
FOV angle		40	50	60	Deg	50% of Maximum

Figure 7. FOV Angle

4.0 Software Reference

4.1 Polling Method Flow Chart (1Hz)

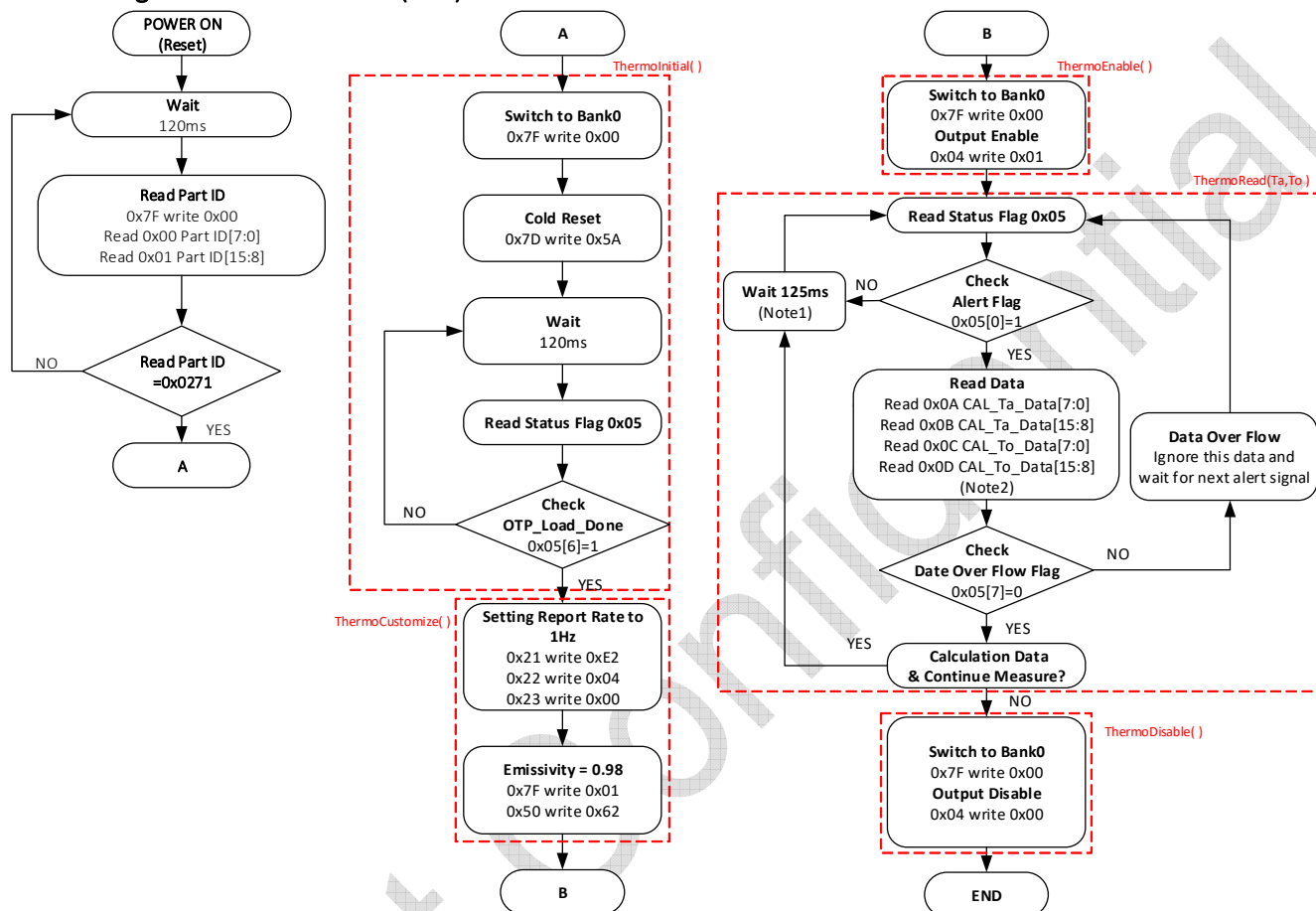


Figure 8. Polling Method Flow Chart

Note 1. Wait time setting base on report rate setting. Wait time $\leq (\text{report time})/2$

Note 2. Conversion data is save to data registers (Bank 0, 0x06 to 0x0D). When host read data register, the data will keep in register until host read Bank0, 0x0D, all alert signals is cleared.

Note 3. Host should be read data within 1 frame time when alert is active (1frame time=1/Report Rate)

4.2 Interrupt Method Flow Chart (1Hz)

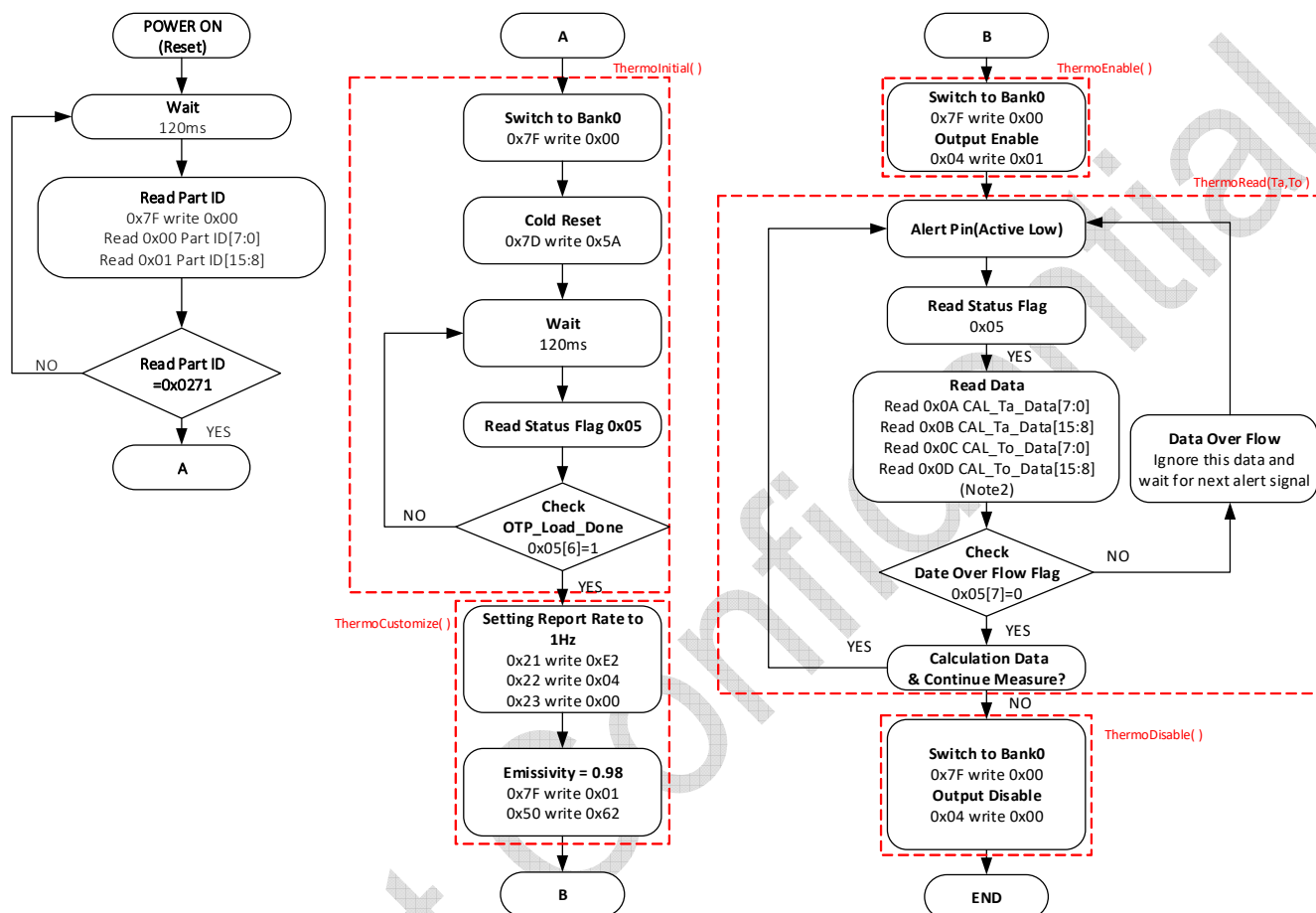


Figure 9. Interrupt Method Flow Chart

Note 2. Conversion data is save to data registers (Bank 0, 0x06 to 0x0D). When host read data register, the data will keep in register until host read Bank0, 0x0D, all alert signals is cleared.

Note 3. Host should be read data within 1 frame time when alert is active (1frame time=1/Report Rate)

5.0 Temperature Data Register and Status Flow

5.1 Temperature Data Register

The sensor can output temperature data and indicator status. The temperature data for T_A and T_O be stored in data registers (Bank 0; address 0x0A to 0x0D).

Address (Bank0)	Register Bit Field	Description
0x0A	CAL_Ta_Data[7:0]	Lower byte of T_A ambient temperature
0x0B	CAL_Ta_Data[15:8]	Upper byte of T_A ambient temperature
0x0C	CAL_To_Data[7:0]	Lower byte of T_O object temperature
0x0D	CAL_To_Data[15:8]	Upper byte of T_O object temperature

The Ambient temperature (*CAL_Ta_Data*) and Object temperature (*CAL_To_Data*) data format is using 16 bits 2's complement. The output scale is 0.03125°C/LSB.

Bit[15]	Bit[14:0]
Sign bit	Ambient temperature= <i>CAL_Ta_Data</i> [14:0] Object temperature= <i>CAL_To_Data</i> [14:0]

For example, read the either *CAL_Ta_Data*[15:0] or *CAL_To_Data*[15:0] to obtain the raw data and check for the sign bit.

- If Bit[15:0] = 0000 0011 1100 0000, then
 - Sign bit = 0
 - CAL_Ta/To_Data*[15:0] = 0000 0011 1100 0000 = 960
 - Effective Temperature = 960 x 0.03125 = 30°C
- If the Bit[15:0] = 1111 1100 0100 0000, then
 - Sign bit = 1
 - CAL_Ta/To_Data*[15:0] – 1 = 1111 1100 0011 1111
 - Bits inverted = 0000 0011 1100 0000 = 960
 - Adding a negative sign to have effective value of *CAL_Ta/To_Data* = -960
 - Effective Temperature = -960 x 0.03125 = -30°C

5.2 Status Register

The status flag is updated for each conversion to indicate sensor status and temperature data event.

Address(Bank0)	Register	Description
0x05 [0]	Alert_flag	Set to 1 if conversion is done
0x05 [6]	OTP_Load_Done	OTP Load Done flag. When calibration coefficients load to system from internal memory is completed, this bit is set to 1.

0x05 [7]	Data_Overflow	This bit is set to 1 if the result in conversion data is over flow and the conversion data is invalid
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5.3 Register Table

There is bank 0 register for accessing the firmware setting to customize for difference usage and application.

Bank 0					
Address	Register Name	Bit	R/W	Initial Value	Description
0x00	PartID[7:0]	[7:0]	R	0x71	Part ID
0x01	PartID[15:8]	[7:0]	R	0x02	
0x04	Output_En	[7:0]	R/W	0x00	0: Output disable 1: Output enable
0x05	Alert_flag	[0]	R	0	Alert flag
	OTP_Load_Done	[6]	R	0	OTP Load Done flag.
	Data_Overflow	[7]	R	0	Data overflow flag. Set by DSP and cleared by read bank0 0x0D
0x0A	CAL_Ta_Data[7:0]	[7:0]	R	0x00	Local ambient sensor temperature data output
0x0B	CAL_Ta_Data[15:8]	[7:0]	R	0x00	
0x0C	CAL_To_Data[7:0]	[7:0]	R	0x00	Object sensor temperature data output
0x0D	CAL_To_Data[15:8]	[7:0]	R	0x00	
0x21	Burst_Freq_Sel[7:0]	[7:0]	R/W	0x38	Report Rate Setting $= (1/20k) \times R_Burst_Freq_Sel[19:0] \times 16$
0x22	Burst_Freq_Sel[15:8]	[7:0]	R/W	0x01	
0x23	Burst_Freq_Sel[19:16]	[3:0]	R/W	0x00	
0x7D	HOST_RSTB[7:0]	[7:0]	W	0x00	0x5A: cold reset, all digital functions will be reset. 0x9A: warm reset, all digital functions will be reset except the register value.

Bank 1					
Address	Register Name	Bit	R/W	Initial Value	Description
0x50	R_EE[6:0]	[6:0]	R/W	0x60	Emissivity compensation. Emissivity % = R_EE[6:0]
0x51	R_VD[7:0]	[7:0]	R/W	0x00	Cover decay compensation. Cover decay % = R_VD[7:0]/2